Alzheimer's Disease Detection from Resting-State EEG Using Explainable CNN Classification and Brain-Age Regression: Feature-Dependent Brain-Age Gaps and External Validation

Aayush Soni, Manasa M, Aamal Magdum, Afreen Anz, and Aparna Sankar

Abstract—Background: Resting-state EEG supports two complementary approaches to Alzheimer's disease (AD): discriminative classification, and normative brain-age regression, whose brain-age gap (BAG) is expected to show "accelerated aging"; alpha-band EEG studies report positive BAG in AD. We developed and evaluated both approaches under an audited, leakage-controlled protocol. Methods: A ridge brain-age model was trained on healthy adults (Dortmund Vital Study, $n = 200$, 20–70 years) and transferred to a clinical cohort (OpenNeuro ds004504: 36 AD, 29 healthy controls [HC], 22 frontotemporal dementia [FTD]). Patients were never used for training, harmonization, bias correction or model selection. We report leave-one-HC-out bias correction, nested model selection, age/sex adjustment, a permutation test that retrains the model ($N = 500$), an external healthy cohort (MPI-LEMON, $n = 70$), a pre-registered feature-family analysis and a multivariate normative-deviation model. A CWT-scalogram CNN and four baselines were evaluated on identical subject-level folds, with nested stacking and occlusion-tested Grad-CAM, and the frozen classifiers were externally validated on an independent Brazilian AD cohort (ADFSU: 80 AD, 11 HC) under a pre-registered plan. Results: The full-spectrum BAG was negative in AD ($d = -0.91$, 95% CI $[-1.50, -0.44]$; age- and sex-adjusted difference -10.3 years, $p = 0.001$; permutation $p = 0.002$). It was absent in FTD ($d = -0.17$) and not shown by healthy older adults from an independent site. The sign depended on the features: alpha-only models made AD look older ($d = 0.65$), slow-wave and aperiodic models made it look younger ($d = -0.97$, -0.84), and 11 of 12 single-family models followed a pre-registered direction prediction (sign test $p = 0.006$). A multivariate normative-deviation score separated AD from HC better than BAG (AUC 0.83 vs. 0.70) but also deviated in FTD. The CNN reached AUC 0.89 and accuracy 82% (three seeds), statistically indistinguishable from spectral baselines and EEGNet. BAG added no information to the CNN in nested stacking ($p = 1.00$). Grad-CAM was stable across seeds, but occlusion did not confirm its emphasis on theta. On the external AD cohort, frozen spectral classifiers transferred well (SVM AUC 0.95, random forest 0.95) and the CNN less well (AUC 0.78). Conclusion: The direction of the EEG brain-age gap in AD is a property of the feature set, not of the disease. AD EEG is better described as a multivariate deviation from normative aging than as accelerated or delayed aging. The classifiers generalize to an independent clinical site; external validation of brain age requires an AD cohort with recorded ages.

Index Terms—Alzheimer's disease, electroencephalography, brain age, normative modeling, continuous wavelet transform, convolutional neural network, data leakage, explainable AI, Grad-CAM, external validation

The authors are with the Department of Computer Science and Engineering, B. M. S. College of Engineering, Bengaluru, India.

I. Introduction

ALZHEIMER'S disease (AD) is the leading cause of dementia. Its diagnosis relies on clinical assessment, neuropsychological testing and, increasingly, MRI, PET or fluid biomarkers. These tools are expensive or invasive and scarce in many health systems. Resting-state EEG is non-invasive, portable and widely available. AD alters it in a characteristic way, often summarized as "slowing": delta and theta activity increase, alpha and beta activity decrease, and the dominant alpha rhythm moves to lower frequencies [1], [2].

Two computational strategies exploit these changes. Discriminative classifiers learn to separate patients from controls. They range from band-power features with classical machine learning to deep networks on raw signals or time–frequency images [3], [4]. Their reported accuracy is inflated when evaluation is not subject-independent [5], [6]. Brain-age models take a normative route. A regressor is trained only on healthy people to predict chronological age, and the brain-age gap (BAG, predicted minus chronological age) of a new person is read as a marker of accelerated or delayed aging [7], [8]. For EEG, positive gaps in dementia have been reported from sleep recordings [9] and, recently, from alpha-band resting-state activity in a large multi-site cohort [10].

The accelerated-aging interpretation assumes that disease moves the EEG along the healthy-aging trajectory. Many spectral features change in healthy aging, however, and AD need not change all of them in the same direction. If it does not, the sign of an EEG brain-age gap would depend on which features the age model uses. Small clinical EEG cohorts are also vulnerable to leakage and analytic flexibility [5], [11].

We therefore asked four questions under an audited, leakage-controlled protocol. (Q1) What is the sign of a full-spectrum resting-state BAG in AD, and does it survive leakage control, covariates and an external healthy site? (Q2) Does the sign depend on the spectral features of the age model, as predicted from feature-wise aging and disease directions? (Q3) Is a multivariate normative-deviation score more informative than a scalar BAG?

(Q4) How does an explainable CWT-scalogram CNN compare with baselines, do BAG or Grad-CAM add reliable information, and do the classifiers generalize to an independent clinical cohort?

Our contributions are:

1) An explainable classification module (CWT-scalogram CNN) benchmarked against four baselines on identical subject folds, with nested stacking, ablations, a three-class (HC/AD/FTD) extension, occlusion-validated Grad-CAM and pre-registered external validation on an independent AD cohort from another country.
2) A normative brain-age regression module whose full-spectrum BAG is robustly negative in AD under extensive leakage control and an external healthy cohort.
3) A pre-registered demonstration that the BAG sign is feature-dependent, which reconciles our result with positive alpha-based gaps.
4) Evidence that a multivariate normative-deviation score outperforms the scalar BAG but is not AD-specific, and that BAG adds interpretability rather than accuracy to the classifier. All code, saved predictions and an application producing per-subject reports are released.

II. Related Work

A. EEG signatures of Alzheimer's disease

Quantitative EEG studies consistently report increased delta and theta power, reduced alpha and beta power, a lower alpha peak frequency and altered connectivity in AD [1], [2]. Babiloni et al. distinguish the pathological slowing of AD from the more modest spectral changes of healthy aging [2].

B. EEG brain age in dementia

Sleep-EEG brain age rises from non-demented to demented groups and correlates with cognitive scores [9]; see also [12]. Otero et al. trained an alpha-only, source-space Elastic Net on healthy adults from ten countries and found older-appearing EEG in MCI, AD and behavioural-variant FTD, with modest effects [10]. Resting-state EEG brain age in healthy populations has been benchmarked [13], [14]. Bias toward the mean age must be corrected [15].

C. EEG classification of AD on ds004504

The dataset authors reported 78.5% accuracy for AD vs. controls with decision trees while comparing validation schemes [16], and 83.3% accuracy for AD vs. controls with leave-one-subject-out validation using DICE-net [17]. EEGNet is a standard compact baseline for raw EEG [18].

D. Leakage, explainability and harmonization

Segment-level splits strongly inflate EEG deep-learning accuracy, including for AD [5], and subject-based cross-validation is required for cross-subject claims [6]. Grad-CAM [19] attributions are widely used on time–frequency images but are rarely tested for faithfulness. ComBat [20] removes site effects while protecting covariates and has been adopted in neuroimaging [21].

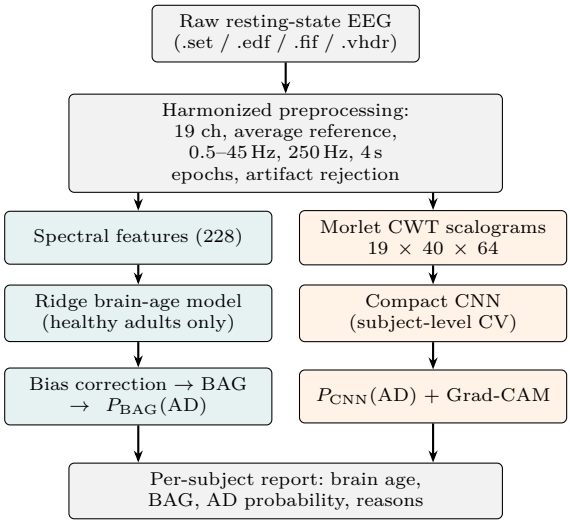


Fig. 1. Proposed pipeline. Teal: normative brain-age regression module. Orange: explainable CNN classifier.

III. Proposed Methodology

Fig. 1 shows the system. A single harmonized preprocessing stage feeds a brain-age regression module (teal) and a CNN classifier (orange); both outputs are reported per subject and can be combined.

A. Harmonized preprocessing

Every recording (training, test or uploaded) passes through one function implemented with MNE-Python [22]. It: (1) maps 10–10 names to 10–20 (T7/T8/P7/P8→T3/T4/T5/T6) and selects 19 channels in a fixed order; (2) re-references to their average; (3) band-pass filters at 0.5–45 Hz (zero-phase FIR); (4) resamples to 250 Hz; (5) cuts 4 s epochs with 2 s overlap; and (6) rejects epochs whose peak-to-peak amplitude on any channel exceeds 150 μV or falls below 1 μV. Recordings shorter than 30 s or with fewer than 10 clean epochs are rejected with an explanatory message. ICA is not used, so the same deterministic pipeline applies at deployment; its effect is tested with ICA-cleaned data. For LEMON, which alternates eyes-open and eyes-closed blocks, only eyes-closed blocks are kept and joined; filtering and epoching never cross the joins.

B. Spectral features

Welch spectra (4 s Hann window, 0.25 Hz resolution) are averaged over clean epochs. Per channel we compute: log absolute and relative power in delta (1–4 Hz), theta (4–8), alpha (8–13) and beta (13–30 Hz); alpha peak frequency ($\arg\max_{f\in[7,13]} P(f)$); log theta/alpha ratio; and aperiodic exponent and offset (FOOOF [23], 2–40 Hz). This yields 12 feature families × 19 channels = 228 features, plus 60 regional averages (5 regions).

C. Brain-age regression module

Model. Ridge regression on standardized features predicts chronological age; the regularization strength is tuned by inner 5-fold CV over $\alpha \in [10^{-2}, 10^{5}]$. A sign-constrained ridge and gradient boosting, and three feature sets, were also evaluated (robustness analysis; Fig. 5).

Transfer and leakage control. Four strategies were compared: (a) healthy cohort only; (b) plus clinical HC, each predicted leave-one-HC-out; (c) ComBat [20], [21] fitted on healthy subjects with age protected; (d) (b)+(c). The strategy is chosen by nested leave-one-HC-out selection. AD and FTD subjects never enter training, harmonization, bias correction or selection; automated guards assert this and that training and test subjects never overlap (src/q1/guards.py; 12 unit tests).

Bias correction and BAG. We fit $\hat{y} = a\,y + b$ on out-of-fold HC predictions excluding the subject being corrected and define $\tilde{y} = (\hat{y} - b)/a$ and $\mathrm{BAG} = \tilde{y} - y$ [15]. A residual correction $(\hat{y} - (ay + b))$ and an uncorrected gap with covariates are reported as alternatives. A logistic model maps BAG to $P_{\mathrm{BAG}}(\mathrm{AD})$; its cross-validation refits the correction within folds.

Statistics. Group effects use Cohen's d (bootstrap CI), Mann–Whitney tests and OLS adjusting for age, sex and epoch count. A permutation test permutes AD/HC labels and retrains the label-dependent model each time ($N = 500$).

D. Feature-family analysis (pre-registered)

After reading [10], and before running the analysis (version-controlled commit), we hypothesized that a single-family age model makes AD look older if the family changes in the same direction in healthy aging and in AD, and younger otherwise: predicted direction $= \mathrm{sign}(r_{\mathrm{age,healthy}} \cdot d_{\mathrm{AD-HC}})$. We trained the same pipeline on each of the 12 families and on alpha, slow-wave and aperiodic sets. Directions use the residual correction, because single-family models can have small or negative bias slopes.

E. Normative-deviation model

For each regional feature a healthy-only reference model (feature $\sim$ age + age^2 + site) is fitted on Dortmund plus clinical HC; each HC is scored by a reference fitted without that subject. Residual z-scores are summarized by a Mahalanobis distance (Ledoit–Wolf covariance).

F. CWT-scalogram CNN classification module

Input. Each clean epoch is transformed with complex Morlet wavelets at 40 log-spaced frequencies (1–40 Hz, $n_{\mathrm{cycles}} = \mathrm{clip}(f/2, 2, 10)$); log power is pooled into 64 time bins, giving a $19 \times 40 \times 64$ tensor with channels as input maps (Fig. 2). Up to 150 evenly spaced epochs are used per subject.

Architecture and training. Three blocks of 3×3 convolution, batch normalization and ReLU (16, 32 and 32 filters; the first two followed by 2×2 max-pooling), global average pooling, dropout 0.5 and a linear output (about 17k parameters). Training uses Adam (learning rate 10^{-3}, weight decay 10^{-3}, batch 128) and class-weighted binary cross-entropy. Inputs are standardized per channel and frequency using training subjects only. Early stopping monitors the loss of validation subjects (15% of the training side; patience 6; at most 40 epochs). A subject's $P_{\mathrm{CNN}}(\mathrm{AD})$ is the mean of its epoch probabilities. The final model is trained on all subjects for 3 epochs (median best epoch in CV).

Explainability. For last-layer activations A^k and logit y^c, Grad-CAM computes $L^c = \mathrm{ReLU}(\sum_k w_k^c A^k)$ with $w_k^c = \frac{1}{Z}\sum_{i,j} \partial y^c / \partial A_{ij}^k$ [19], upsampled to the 40×64 grid. Band importance is the mean map value within a band divided by the overall mean (1.0 = average). Channel importance uses gradient×input. Faithfulness is tested by occlusion: each fold model re-scores its untouched test subjects with one band or scalp region set to the training mean.

TABLE I
Cohorts after preprocessing

Cohort	n	Age, mean ± SD (range)	Female	MMSE
ds004504 AD	36	66.4 ± 7.9 (49–79)	24/36	17.8 ± 4.5
ds004504 HC	29	67.9 ± 5.4 (57–78)	11/29	30.0 ± 0.0
ds004504 FTD	22	64.3 ± 7.8 (44–78)	9/22	22.0 ± 2.6
Dortmund (HC)	200	49.7 ± 14.0 (22–70)	121/200	–
LEMON (ext. HC)	70	55–80 (5-y bins)	–	–

G. Baselines, combination and evaluation

On the CNN's identical subject folds we evaluated logistic regression, RBF-SVM and random forest on the 228 spectral features (inner 5-fold CV tuning) and EEGNet-8,2 [18] on raw epochs (same training loop). CNN and BAG were combined by stacking, evaluated fully nested (Algorithm 4). All metrics are subject-level and out-of-fold: AUC, PR-AUC, accuracy, balanced accuracy, precision, sensitivity, specificity, F1, Brier score and expected calibration error, with 2000-sample stratified bootstrap CIs; Holm-corrected paired-bootstrap comparisons; and a CNN permutation test that re-runs the full CV [24]. The CNN was run with three seeds, each changing fold assignment and initialization.

IV. Algorithms

Algorithms 1–5 formalize the pipeline. Algorithm 2 guarantees that no patient influences the age model or its bias correction, and that no subject is corrected with its own data. Algorithms 3 and 4 guarantee that normalization, early stopping, meta-learning and testing never share subjects. Inference takes about 10 s per recording on a laptop CPU.

V. Dataset

Table I summarizes the cohorts; all are public and de-identified.

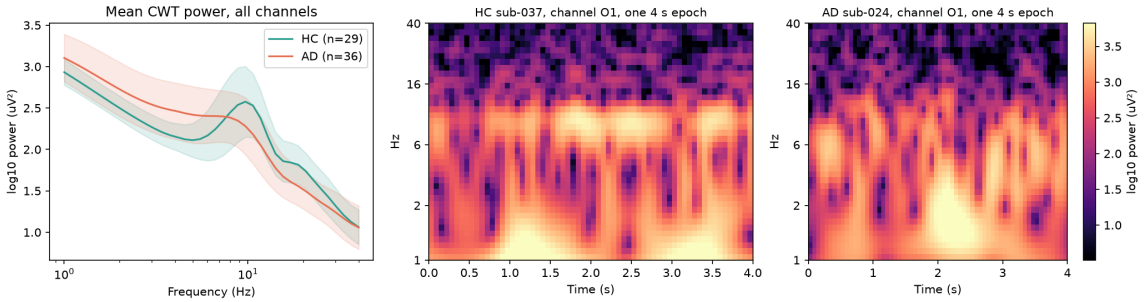


Fig. 2. CWT scalograms. Left: mean power spectra (HC vs. AD). Middle/right: one 4 s epoch at O1 for a healthy and an AD subject.

Algorithm 1 Harmonized preprocessing

Require: EEG recording R; channel list $\mathcal{C}$ (19, 10–20)
Ensure: clean epochs $X \in \mathbb{R}^{n\times 19\times 1000}$

map 10–10 names (T7,T8,P7,P8) to 10–20 (T3,T4,T5,T6)
if a channel of $\mathcal{C}$ is missing or duration < 30 s then
    reject with a message
end if
optionally keep only eyes-closed blocks (alternating protocols)
select $\mathcal{C}$ in fixed order; re-reference to their average
band-pass 0.5–45 Hz (zero-phase FIR); resample to 250 Hz
cut 4 s epochs, 2 s overlap (never across block boundaries)
keep epoch iff $1\,\mu\text{V} < \text{ptp}_c < 150\,\mu\text{V}$ for every channel c
if fewer than 10 epochs remain then
    reject
end if
return X (cached by configuration hash)

Algorithm 2 Brain-age training, transfer and leakage-free BAG

Require: healthy cohort D_H; clinical HC T_C; patients T_P (AD, FTD)
Ensure: BAG for every clinical subject

for each $s \in T_C$ do ▷ leave-one-HC-out
    fit ridge (inner-CV α) on $D_H \cup T_C \setminus \{s\}$; $\hat{y}_s \leftarrow$ prediction
end for
fit ridge on $D_H \cup T_C$; $\hat{y}_p \leftarrow$ prediction for each $p \in T_P$
for each clinical subject i do
    fit $\hat{y} = a\,y + b$ on HC predictions excluding i
    $\text{BAG}_i \leftarrow (\hat{y}_i - b)/a - y_i$
end for
return BAG; fit $P(\text{AD} \mid \text{BAG})$ by logistic regression

Algorithm 3 Subject-level cross-validation of the CNN

Require: subjects S with labels; scalograms Z_s (≤ 150 per subject); seed r
Ensure: out-of-fold $P(\text{AD})$ for every subject

split S into 10 stratified folds (seed r)
for $k = 1$ to 10 do
    $S_{\text{tr}} \leftarrow S \setminus F_k$; split into fit (85%)/validation (15%) subjects
    standardize with fit-subject statistics; train CNN on fit epochs
    early-stop on validation-subject loss (patience 6)
    for each $s \in F_k$ do
        $P_s \leftarrow$ mean epoch probability of s
    end for
end for
return $\{P_s\}$ ▷ no subject in both training and test

Algorithm 4 Fully nested stacking (outer fold k)

Require: outer-train subjects T, outer-test subjects E (disjoint)

inner 5-fold CV of the CNN within $T \rightarrow$ out-of-fold P_{CNN} for T
train the CNN on $T \rightarrow P_{\text{CNN}}$ for E
age model on Dortmund + HC$\in T$ (HC leave-one-out); E never trained on
bias correction from HC$\in T$; BAG for T and E
logistic meta-model on [logit P_{CNN}, BAG] of T; predict E

Clinical cohort: OpenNeuro ds004504 [25]. Routine clinical eyes-closed EEG, 19 channels (10–20), 500 Hz, with MMSE scores. One FTD recording failed artifact screening. AD and HC did not differ in age ($p = 0.38$) but differed in sex (AD 67% vs. HC 38% female, $p = 0.04$) and number of clean epochs ($p = 0.02$); both are used as covariates. ICA-cleaned derivatives were used only for sensitivity analysis. There are no MCI subjects.

Healthy training cohort: OpenNeuro ds005385, Dortmund Vital Study [26]. 64-channel EEG at 1000 Hz from healthy adults (20–70 years); baseline eyes-closed pre-task recordings; recordings flagged as possibly discontinuous excluded; age-stratified sample weighted toward older ages ($n = 200$). A replication used all eligible subjects ($n = 513$ after preprocessing).

External healthy cohort: MPI-LEMON. 62-channel EEG at 2500 Hz; older group (5-year age bins, 55–80; bin midpoints used, ±2.5 years label noise); eyes-closed blocks only; $n = 70$ of 71 available. Never used to train the main

Algorithm 5 Inference and report for a new subject

Require: EEG file, chronological age y

1: $X \leftarrow$ Algorithm 1(file)
2: $\hat{y} \leftarrow$ ridge(features(X)); BAG $\leftarrow (\hat{y} - b)/a - y$
3: $P_{\mathrm{CNN}} \leftarrow$ mean CNN probability over scalograms of X
4: Grad-CAM band importance toward the predicted class
5: compare markers and BAG with HC/AD reference values
6: return brain age, BAG, P_{CNN}, P_{BAG}, combined P, reasons

model; its analysis plan was committed before processing.

External clinical cohort: ADFSU [27]. Resting-state EEG from Brazil (OSF 2v5md): 80 AD patients (NINCDS-ADRDA and DSM-III-R criteria) and 12 HC, 19 channels (10–20), 128 Hz, 8 s segments band-limited to 0.5–30 Hz by the data providers, eyes-closed and eyes-open conditions. Ages and sex are not provided per subject, so the cohort can test the classifiers but not brain age. Recordings passed through the same preprocessing; only the minimum-length rule was relaxed to accept 8 s recordings (at least 2 clean 4 s epochs). The cohort was never used for training, tuning or threshold selection, and the analysis plan was committed before any prediction.

Data quality. On average about 80% of epochs were retained for clinical subjects and 90% for the healthy cohort; median peak-to-peak amplitudes were similar across cohorts, confirming consistent units and scaling.

VI. Results

A. Q1: The full-spectrum BAG is negative in AD

Age prediction. On the healthy cohort the model reached MAE 9.2 years ($r = 0.60$) in 10-fold CV, with the expected regression to the mean (mean signed error 13.8 years in the 20s, −7.2 in the 60s, −15.8 at 70). On clinical HC the out-of-fold MAE was 7.4 years ($r = 0.53$; bias slope $a = 0.88$). R^2 was −1.9, which is uninformative for a 57–78-year range, so it is not used as a success criterion.

Group differences. With leakage-free correction the BAG was 0.0 ± 9.4 years in HC, -11.2 ± 14.2 in AD and −2.2 in FTD (Fig. 3). AD differed from HC ($d = -0.91$, [−1.50, −0.44], Mann–Whitney $p < 0.001$) and from FTD ($d = -0.57$, $p = 0.04$); FTD did not differ from HC ($d = -0.17$, $p = 0.93$). The leakage corrections changed the effect minimally (Table II, Fig. 4; originally reported $d = -0.93$).

Covariates and sensitivity. Adjusting for age and sex gave −10.3 years (SE 3.0, $p = 0.001$); adding epoch count gave −9.2 years ($p = 0.003$). Residual and uncorrected gaps gave −9.1 and −9.1 years ($p < 0.001$). The retrained permutation test gave $p = 0.002$ (null SD of d 0.28). All 12 numerically stable configurations were negative (d −0.94 to −0.27); 7 remained significant after age/sex adjustment (Fig. 5). The effect persisted on ICA-cleaned recordings ($d = -0.70$, $p = 0.02$); with 513 healthy training subjects ($d = -0.83$; healthy CV MAE 8.4 years, $r = 0.71$, but HC MAE 12.6 years; AUC 0.71; permutation $p = 0.003$; AD vs. FTD $p = 0.002$); and across artifact thresholds of 100–200 μV (d −0.91 to −0.64; all age/sex-adjusted $p \leq p = 0.02$). The magnitude was threshold-sensitive: the pre-specified 150 μV threshold gave the largest effect.

TABLE II
AD−HC brain-age effect under the leakage and bias-correction audit

Analysis	Cohen's d [95% CI]	MWU
As originally reported (in-sample HC bias correction)	−0.93 [−1.51, −0.47]	$p < 0.001$
Leave-one-HC-out bias correction	−0.91 [−1.50, −0.44]	$p < 0.001$
+ nested strategy selection	−0.91 [−1.50, −0.44]	$p < 0.001$
Residual bias correction	−0.92 [−1.50, −0.46]	$p < 0.001$
No bias correction	−0.89 [−1.47, −0.44]	$p = 0.001$

Sex. Within HC, women had lower BAG than men (−5.5 vs. 3.5 years, $p = 0.01$). Stratified by sex, the AD effect was present in men ($d = -1.00$, $p = 0.02$; 12 AD vs. 18 HC) and in the same direction but not significant in women ($d = -0.58$, $p = 0.12$; 24 vs. 11).

External healthy site (LEMON). Test L1 (unchanged model): predictions for 70 LEMON adults had MAE 10.7 years and did not track age within 55–80 years ($r = -0.03$). Uncorrected gaps were -7.8 ± 11.5 years in LEMON HC, -5.0 ± 7.7 in clinical HC and -14.7 ± 12.9 in AD (Fig. 6). Healthy adults from the two sites did not differ ($d = 0.26$, $p = 0.18$), whereas AD differed from the external healthy cohort ($d = -0.58$, $p = 0.007$), arguing against a site or age artifact. Test L2 (retraining on healthy cohorts spanning 20–80 years, $n = 270$, healthy CV MAE 9.1 years) preserved the effect ($d = -0.60$, $p = 0.01$; AUC 0.63; FTD $d = -0.11$), negative in 6 of 6 stable configurations, although HC transfer was weaker (MAE 8.1 years).

Contribution of this result: a leakage-controlled demonstration that the full-spectrum EEG brain-age gap is negative in AD, robust to covariates, preprocessing and an external healthy site, contrary to the expected "accelerated aging".

Clinical association and classification. BAG was not associated with MMSE in AD ($\rho = -0.08$, $p = 0.62$; Fig. 7). As a single-feature classifier, BAG gave AUC 0.70, accuracy 66%, precision 68%, recall 72%, specificity 59% and F1 0.70 (decision rule BAG < -2.6 years).

B. Q2: The sign of the BAG depends on the features

Table III and Fig. 8 contrast each family's correlation with age in healthy adults with its AD−HC effect. Alpha peak frequency, alpha power and the theta/alpha ratio change in the same direction in aging and AD, whereas beta power rises with age but falls in AD, and slow-wave power and the aperiodic component fall with age but rise in AD. Consistent with this, 11 of 12 single-family brain-age models followed the pre-registered direction (Table IV, Fig. 9; sign test $p = 0.006$). An alpha-only

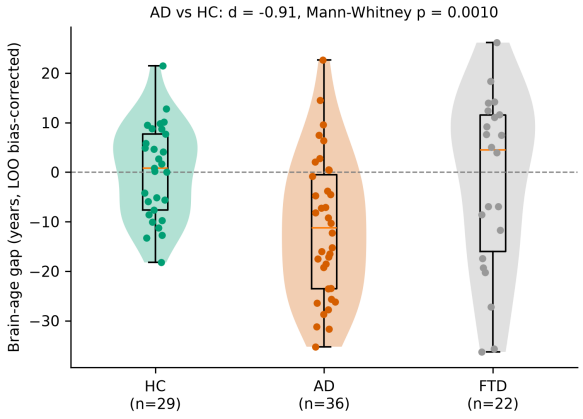


Fig. 3. Leakage-free brain-age gap by group (HC out-of-fold with leave-one-out bias correction; AD and FTD never trained on).

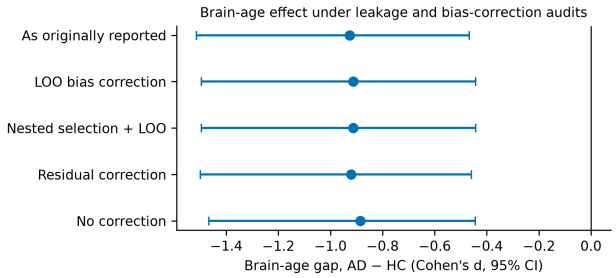


Fig. 4. Brain-age effect (AD−HC) under leakage and bias-correction variants.

model made AD look older ($d = 0.65$, $p = 0.002$), qualitatively reproducing [10]; slow-wave and aperiodic models made AD look younger ($d = -0.97$, -0.84). The full-spectrum model, dominated by these families, was negative ($d = -0.92$). The single mismatch, the theta/alpha ratio, was not significant.

Contribution of this result: a pre-registered explanation of why EEG brain-age studies disagree: the sign of the gap is determined by the feature families of the age model.

C. Q3: Normative deviation outperforms the scalar BAG

The multivariate deviation score separated AD from HC with $d = 1.37$ [0.94, 1.95] and AUC 0.83 (cross-validated 0.83; age/sex-adjusted $p < 0.001$), vs. BAG AUC 0.70. It was not AD-specific: FTD also deviated ($d = 1.09$, $p < 0.001$) and AD did not differ from FTD ($p = 0.36$). AD showed elevated slow-wave and aperiodic z-scores and reduced alpha, beta and alpha-peak z-scores (Fig. 10).

Contribution of this result: a multivariate normative-deviation score is a more informative regression-based marker than the scalar BAG, although not AD-specific.

D. Q4a: CNN classification and baselines

Table V reports the CNN over three seeds: mean AUC 0.89 (range 0.86–0.90), accuracy 82%, precision 84%, recall 83%, specificity 80%, F1 0.84. For the primary seed: AUC 0.86 [0.76, 0.94], PR-AUC 0.89, balanced accuracy 0.80, Brier 0.16, ECE 0.10 (Fig. 13); permutation $p = 0.010$ ($N = 100$). With the originally used early-stopping criterion (validation AUC; Section VI-I) the AUC was 0.86 (accuracy 78%). On identical folds EEGNet (0.88), random forest (0.86), SVM (0.85) and logistic regression (0.84) were statistically indistinguishable from the CNN (Holm-adjusted $p = 1.00$ for every comparison; Table VI, Figs. 11 and 12). Performance was similar for women and men (AUC 0.86 vs. 0.89) and across age tertiles (0.77–0.89).

Contribution of this result: an explainable CNN benchmarked fairly (identical subject folds, calibration, permutation testing) against four baselines, without inflated claims of superiority.

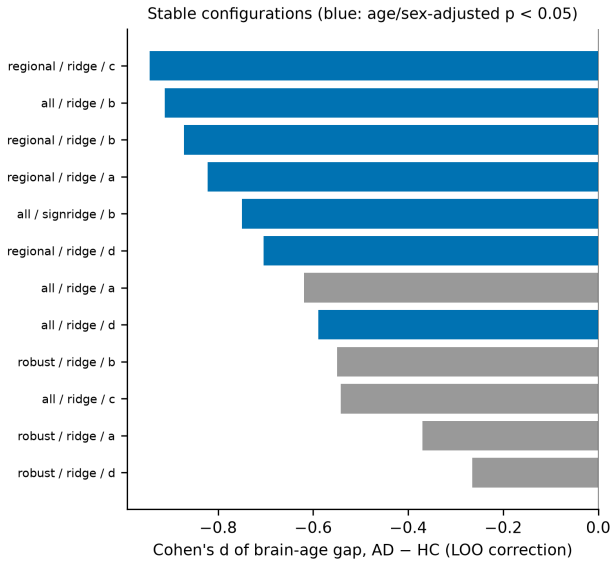


Fig. 5. AD−HC effect in every numerically stable configuration (feature set / model / strategy); blue = age/sex-adjusted $p < 0.05$.

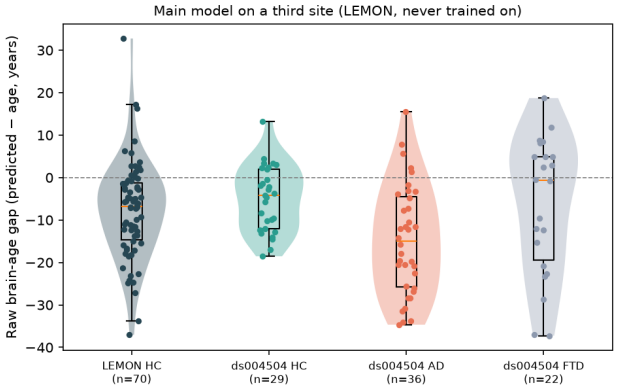


Fig. 6. Uncorrected brain-age gaps from the unchanged main model on an external healthy cohort (LEMON) and the clinical cohort.

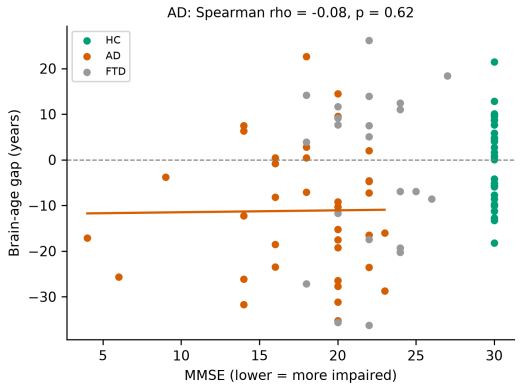


Fig. 7. Brain-age gap vs. MMSE (no association within AD).

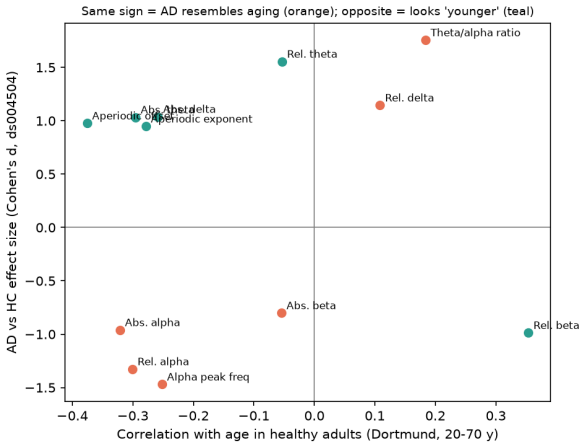


Fig. 8. Healthy-aging direction (x) vs. AD effect (y) per feature family.

TABLE III
Healthy-aging direction vs. AD effect per feature family

Feature	r (age, healthy)	d (AD−HC)	AD looks
Alpha peak freq	−0.25	−1.47	older
Rel. alpha	−0.30	−1.33	older
Abs. alpha	−0.32	−0.96	older
Theta/alpha ratio	+0.18	+1.75	older
Rel. theta	−0.05	+1.55	younger
Abs. theta	−0.29	+1.03	younger
Rel. delta	+0.11	+1.15	older
Abs. delta	−0.26	+1.04	younger
Rel. beta	+0.35	−0.99	younger
Abs. beta	−0.06	−0.80	older
Aperiodic exponent	−0.28	+0.95	younger
Aperiodic offset	−0.38	+0.98	younger

TABLE IV
Single-family brain-age models: predicted vs. observed AD direction

Feature family	Predicted	d (AD−HC)	p	Agree
Absolute delta	younger	−0.97	$p < 0.001$	yes
Absolute theta	younger	−0.93	$p < 0.001$	yes
Aperiodic offset	younger	−0.87	$p = 0.001$	yes
Relative beta	younger	−0.84	$p = 0.002$	yes
Aperiodic exponent	younger	−0.74	$p = 0.007$	yes
Relative theta	younger	−0.71	$p = 0.005$	yes
Theta/alpha ratio	older	−0.34	$p = 0.20$	no
Relative delta	older	0.29	$p = 0.14$	yes
Relative alpha	older	0.38	$p = 0.02$	yes
Absolute beta	older	0.82	$p = 0.006$	yes
Absolute alpha	older	0.90	$p < 0.001$	yes
Alpha peak frequency	older	1.23	$p < 0.001$	yes

E. Q4b: Combining classification and regression

In fully nested stacking the CNN alone gave AUC 0.87 (accuracy 78%), BAG alone 0.72 (vs. chance $p = 0.002$) and CNN+BAG 0.87 (accuracy 74%; $\Delta = 0.001$, [−0.07, 0.07], $p = 1.00$). A simpler repeated-CV stacking gave the same conclusion (0.87 vs. 0.86, $p = 0.74$). BAG therefore adds interpretability, not accuracy.

Contribution of this result: a fully nested test of whether brain age adds to an EEG AD classifier, showing that the two methods are complementary in interpretation rather than in accuracy.

F. Q4c: Ablations and three-class classification

Ablations (Table VII; three seeds each) gave mean AUCs from 0.84 to 0.89: 8 channels 0.85, 40% of epochs 0.88, only 8–40 Hz 0.87, only 1–8 Hz 0.85, no time axis 0.84, suggesting that low-density montages and shorter recordings are viable. For HC/AD/FTD the CNN reached balanced accuracy 63% (macro AUC 0.78, macro F1 0.63) vs. 56% (0.72) for a random forest; recall was 76% (HC), 69% (AD) and 45% (FTD), with 10 of 22 FTD subjects labelled AD (Table VIII).

G. Q4d: Explainability and its validation

Final-model Grad-CAM band importance toward AD was theta 1.29, alpha 1.39, beta 1.07, delta 0.83; toward HC alpha 1.37 and beta 1.40 (Fig. 14); the top AD channels were F8, T6, T3, P4 (Fig. 15). Out-of-fold Grad-CAM was stable across seeds (r 0.94–0.99; theta 1.25, alpha 1.29, beta 1.15; Fig. 16). Occlusion, however, showed that the models depended mostly on beta (AUC drop 0.108) and delta (0.059), little on alpha (0.018) and not on theta (0.002); spatially, occipital (0.124), frontal (0.114) and temporal (0.102) regions mattered most (Fig. 17). Grad-CAM importance therefore did not identify the information the model needs.

Contribution of this result: evidence that Grad-CAM explanations of EEG classifiers must be validated by perturbation; stability across seeds is not faithfulness.

H. Q4e: External clinical validation (ADFSU)

In the primary analysis (E1) the final models trained only on ds004504 were applied unchanged to ADFSU eyes-closed recordings (80 AD, 11 HC; one HC excluded by the artifact rule). With the fixed 0.5 decision threshold, the RBF-SVM reached AUC 0.95 [0.90, 0.99], balanced accuracy 86% (sensitivity 99%, specificity 73%); the random forest 0.95 [0.87, 1.00]; logistic regression 0.89 [0.75, 0.99]; and the CNN 0.78 [0.62, 0.91] (balanced accuracy 68%, sensitivity 54%, specificity 82%) (Table IX). Retraining on ds004504 after the same 30 Hz low-pass

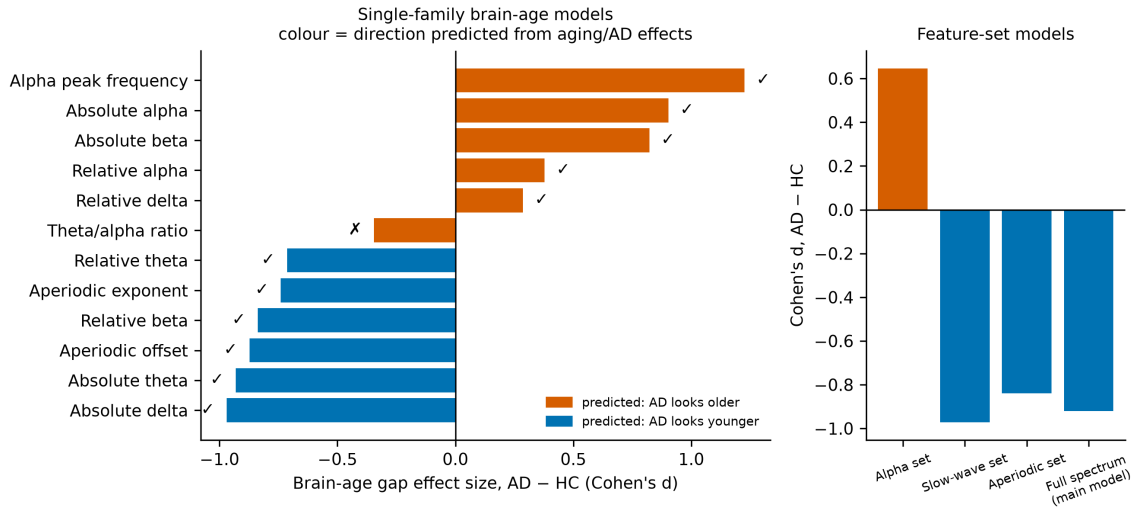


Fig. 9. Feature-dependence of the brain-age gap. Left: single-family models coloured by the direction predicted before the analysis (✓ = observed direction agrees). Right: feature-set models.

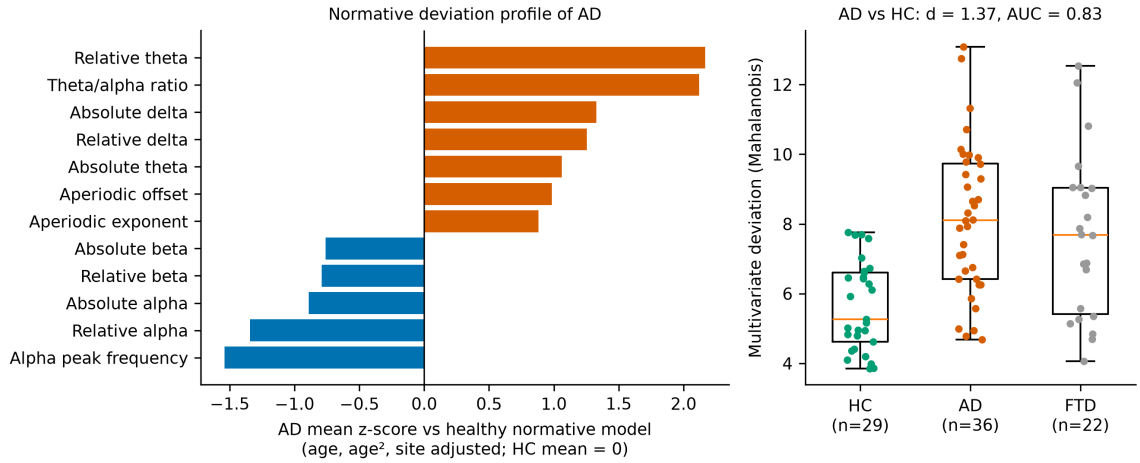


Fig. 10. Normative deviation. Left: AD mean z per family relative to an age-, age^2- and site-adjusted healthy reference. Right: multivariate deviation by group (HC scored leave-one-out).

TABLE V
CNN, AD vs. HC, subject-level 10-fold CV per seed ($n = 65$)

Seed	AUC	Acc.	Prec.	Rec.	Spec.	F1
42	0.86	80	81	83	76	0.82
7	0.90	85	88	83	86	0.86
123	0.90	82	83	83	79	0.83
Mean ± SD	0.89 ± 0.03	82 ± 2	84 ± 4	83 ± 0	80 ± 5	0.84 ± 0.02

(E2) did not change the ranking (CNN 0.78, SVM 0.93, random forest 0.94, logistic regression 0.88). On eyes-open recordings (76 AD, 11 HC), which the models never saw during training, AUCs were lower (SVM 0.82, CNN 0.72). Spectral models thus transferred better across site, country, amplifier and sampling rate than the CNN, whose time–frequency images are more sensitive to recording length and bandwidth. With only 11 HC, the CIs are wide.

Contribution of this result: a pre-registered external validation of ds004504-trained AD classifiers on an independent cohort from another country, showing that subject-independent spectral classifiers generalize, and that internal parity between CNN and spectral models does not imply equal transferability.

I. Discussion

Why the sign depends on the features. Healthy aging and AD share alpha slowing but move absolute slow-wave power, beta power and the aperiodic component in opposite directions [2]. A brain-age model projects the EEG onto the healthy-aging axis; its sign in AD is determined by which features dominate that projection. Our pre-registered family analysis confirmed this (11 of 12), and an alpha-only model reproduced the positive

TABLE VI
AD vs. HC, out-of-fold on identical subject folds ($n = 65$), with 95% bootstrap CIs for AUC

Model	AUC [95% CI]	PR-AUC	Bal. acc.	Sens.	Spec.	Brier
CWT-scalogram CNN	0.86 [0.76, 0.94]	0.89	0.80	0.83	0.76	0.16
EEGNet	0.88 [0.80, 0.95]	0.92	0.80	0.81	0.79	0.14
Random forest	0.86 [0.76, 0.94]	0.92	0.77	0.75	0.79	0.14
RBF-SVM	0.85 [0.74, 0.94]	0.90	0.81	0.75	0.86	0.15
Logistic regression	0.84 [0.73, 0.93]	0.89	0.76	0.72	0.79	0.17
Brain-age gap	0.73 [0.61, 0.85]	0.83	0.65	0.72	0.59	0.21
CNN + BAG (stacking)	0.87 [0.78, 0.95]	0.92	0.78	0.81	0.76	0.14

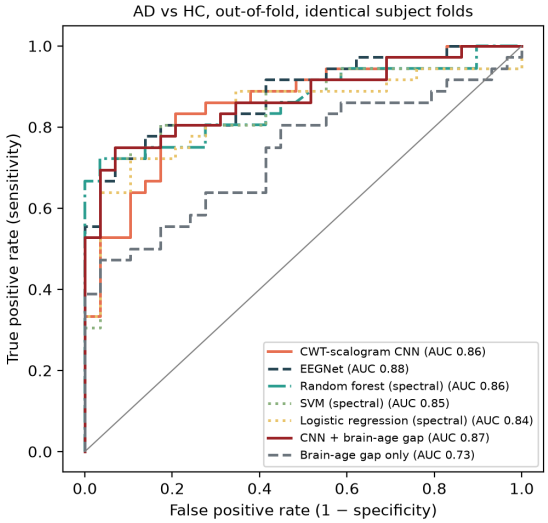


Fig. 11. Out-of-fold ROC curves on identical subject folds.

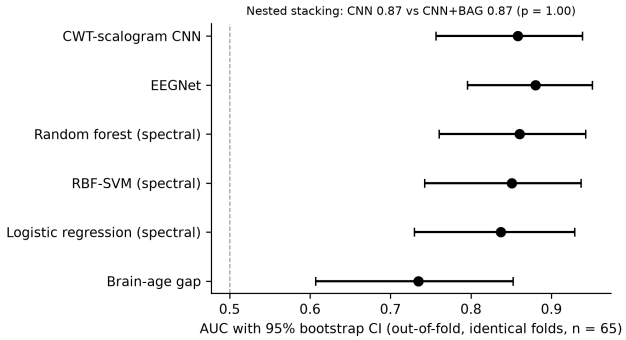


Fig. 12. AUC with 95% CIs on identical folds; nested-stacking comparison in the title.

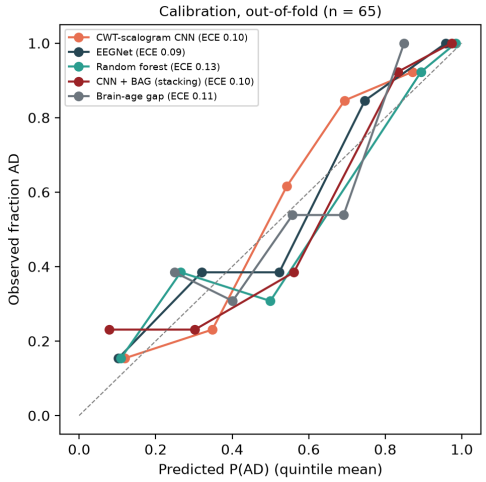


Fig. 13. Calibration of out-of-fold probabilities (quintiles).

TABLE VII
CNN ablations (mean ± SD over seeds)

Variant	AUC	Acc. (%)	seeds
Full model (19 ch, 1–40 Hz, time)	0.89 ± 0.03	82 ± 2	3
Only 1–8 Hz	0.85 ± 0.01	77 ± 4	3
Only 8–40 Hz	0.87 ± 0.02	79 ± 2	3
No time axis (spectrum)	0.84 ± 0.04	79 ± 5	3
8 channels	0.85 ± 0.02	79 ± 3	3
40% of epochs	0.88 ± 0.03	78 ± 2	3

gaps of [10]. Positive and negative EEG BAGs in AD are therefore not contradictory findings but different projections of the same deviation. The multivariate normative score, which does not force the deviation onto an age axis, captured it better (AUC 0.83), supporting normative-deviation reporting over a "brain age".

Alternative explanations. We controlled for leakage, bias correction, model choice, sex, age, epoch count, artifact threshold, ICA and training age range; the effect remained, and healthy older adults from an independent site did not show it. Because BAG is not associated with MMSE and FTD shows a similar multivariate deviation, we do not claim BAG as a severity or AD-specific biomarker. Sex is a relevant moderator (women appeared younger within HC); residual confounding cannot be excluded with $n = 65$.

Classification and explanations. The CNN performed comparably to tuned spectral baselines, EEGNet and published subject-independent results on this dataset [16], [17]; we do not claim superiority. Grad-CAM was stable but not faithful: occluding theta, emphasized by Grad-CAM, did not reduce performance, consistent with redundancy across bands. Stability of explanations should not be mistaken for validity.

Transparency. The CNN early-stopping criterion was changed from validation AUC to validation loss after the first run's outer-fold output had been seen (both reported above); the sign-constrained and regional brain-age variants and an alpha-only model were added after

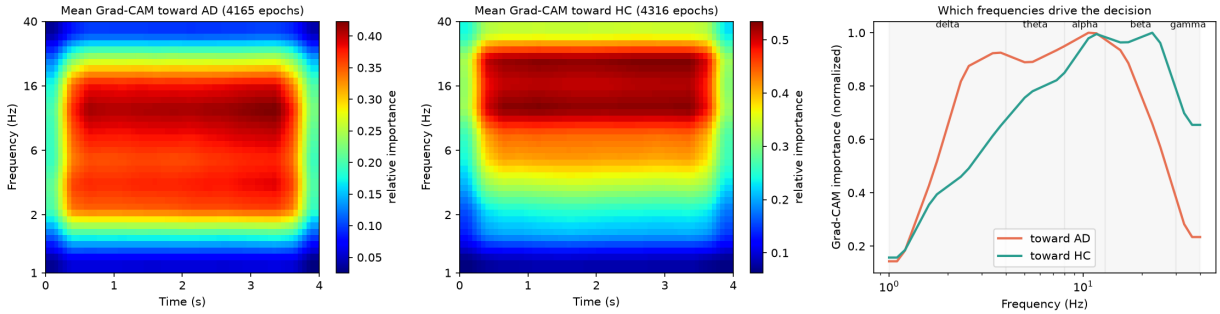


Fig. 14. Grad-CAM of the final CNN: mean maps toward AD and HC and frequency profiles.

TABLE VIII
Three-class CNN confusion matrix (primary seed; rows: true class)

True \ predicted	HC	AD	FTD
HC	22	3	4
AD	6	25	5
FTD	2	10	10

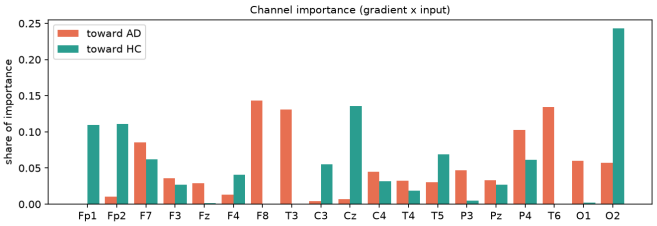


Fig. 15. Channel importance (gradient×input) of the final CNN.

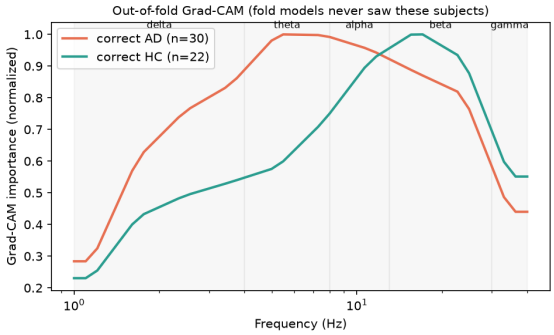


Fig. 16. Out-of-fold Grad-CAM profiles for correctly classified AD and HC subjects.

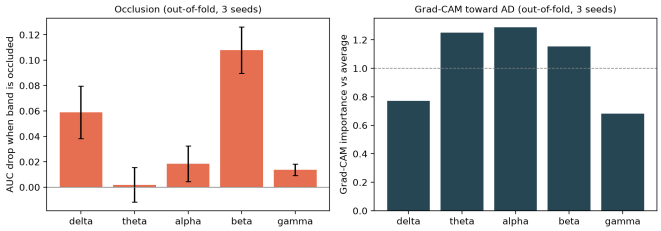


Fig. 17. Explanation validation: AUC drop when a band is occluded (left) vs. Grad-CAM band importance (right), out-of-fold, three seeds.

the main result. All such decisions are disclosed in the repository's leakage audit.

VII. Conclusion and Contribution

This work detects AD from resting-state EEG with two complementary methods, classification and brain-age regression, and contributes:

1) Explainable classification. A CWT-scalogram CNN reached AUC 0.89 and accuracy 82%, statistically indistinguishable from spectral baselines and EEGNet; ablations and a three-class extension are reported, and Grad-CAM explanations were found stable but not faithful under occlusion.

2) External clinical validation. Under a pre-registered plan, frozen classifiers generalized to an independent Brazilian AD cohort (SVM AUC 0.95, random forest 0.95, CNN 0.78), showing that spectral models transfer better than the CNN.

3) Normative brain-age regression. A full-spectrum brain-age model trained only on healthy adults places AD younger ($d = -0.91$, $[-1.50, -0.44]$), robustly to leakage control, covariates, preprocessing, training-set size and age coverage, and not reproduced by healthy older adults from an independent site.

4) Feature-dependent sign. The direction of the gap is a property of the feature set (11 of 12 pre-registered predictions; alpha-only models place AD older), reconciling conflicting reports; AD EEG is better described as a multivariate normative deviation (AUC 0.83).

5) Combination. BAG adds interpretability, not accuracy, to the classifier (nested $p = 1.00$); both are delivered by an open application producing per-subject reports.

Limitations. The development cohort is small and single-site. The external AD cohort validates the classifiers only: it has no ages (so brain age could not be validated externally), only 11 usable HC, 8 s recordings and a 30 Hz band limit. MMSE is the only clinical scale and there are no MCI subjects. LEMON ages are binned. Our features are sensor-space and average-referenced over 19 channels, whereas [10] used source-space alpha; the qualitative agreement of our alpha-only model does not establish quantitative equivalence. The effect size depends on the artifact threshold.

TABLE IX
External validation on ADFSU (eyes closed, 80 AD vs. 11 HC). Internal: out-of-fold AUC on ds004504; E1: frozen models; E2: retrained after 30 Hz low-pass; EO: E1 on eyes-open

Model	Internal	E1 AUC [95% CI]	E1 BA (%)	E2	EO
CWT-scalogram CNN	0.86	0.78 [0.62, 0.91]	68	0.78	0.72
RBF-SVM (spectral)	0.85	0.95 [0.90, 0.99]	86	0.93	0.82
Random forest (spectral)	0.86	0.95 [0.87, 1.00]	85	0.94	0.77
Logistic regression (spectral)	0.84	0.89 [0.75, 0.99]	78	0.88	0.70

Future work. Brain-age validation on larger independent clinical cohorts with ages, AD and MCI (e.g., CAUEEG, BrainLat); source-space and connectivity features within the same family framework; lifespan normative models of EEG spectra; longitudinal prediction; and prospective evaluation with low-density, short recordings.

Data and Code Availability

OpenNeuro ds004504 and ds005385, MPI-LEMON and ADFSU (OSF 2v5md) are public. All code, saved models, out-of-fold predictions, fold assignments, seeds, audit reports and the scripts generating every number in this paper are in the project repository.

Ethics Statement

Only publicly available, de-identified data collected under the original studies' approvals were used. No new data were collected.

Declaration of Generative AI Use

During the preparation of this work the authors used Claude (Anthropic), an AI assistant, to help write and debug the analysis code and to draft and edit the manuscript text. The authors defined the research questions and study design, reviewed and verified all code, analyses and results, checked every reference against bibliographic databases, and take full responsibility for the content of this publication.

Author Contributions

A. Soni, A. Magdum, A. Anz and A. Sankar contributed equally to conceptualization, methodology, software, validation, formal analysis, investigation, data curation, visualization, and writing (original draft). Manasa M supervised the project and contributed to methodology and to reviewing and editing the manuscript. All authors read and approved the final manuscript.

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This research received no specific grant from any funding agency in the public, commercial, or not-for-profit sectors. The authors declare no conflict of interest.

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